

PHYSICAL · CO-OP · CHAOS

UNDERCOOKED IRL

Overcooked, rebuilt with real appliances, real
RFID tags, and a laptop that keeps score.

undercooked-irl

[Team / hackathon name]



GAMEPLAY LOOP
9:16 · loops continuously

MOTIVATION

Why We Built This

Overcooked is one of the best co-op games ever made — and it's still just a screen. Everyone's crowded around a TV, mashing buttons, yelling at pixels. We wanted the yelling to be about a real frying pan.

The itch — every time we played, someone said “imagine if this was real.” Eventually we stopped imagining.

The gap — no off-the-shelf kit turns a living room into a physical co-op kitchen, so we built the kit ourselves.

[Your story] — *swap this card for your team's actual origin story — a party, a bet, a class project.*



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THE CONCEPT

Overcooked, Played With Real Appliances

Up to eight physical cooking stations, cheap RFID tags standing in for ingredients, and a laptop that runs the whole game — the chaos of the video game, except you're the one holding the joystick and the frying pan.

In the video game	In our kitchen
Press a key to chop	Press a real limit switch on a real cutting board
Waggle a stick to fry	Waggle a joystick wired straight into an ESP32
Watch a countdown on screen	Watch the same countdown on a browser everyone can see
Reset with Ctrl+R	Reset with one button — the laptop owns every bit of state



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LIVE DEMO

See It In Action

PHOTO

Full six-station
table setup

PHOTO

Inside a station —
wiring & RFID reader

VIDEO

30-second
gameplay clip

QR

Scan for a live gameplay clip.

[Swap in real photos, a short video, and a QR code linking to it before presenting.]



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HOW IT PLAYS

One Round, Start To Finish



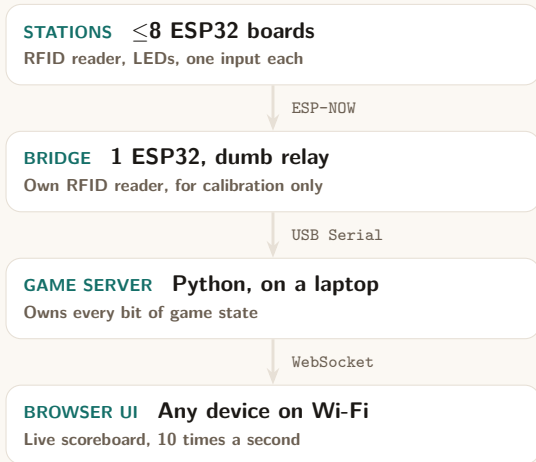
Progress lives on the server, not the station — pick food up mid-task, put it down on a different board of the same kind, and pick up exactly where you left off.



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Four Hops, One Source Of Truth



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Reset the round with one function call — every station just draws what it's told.

CORE PRINCIPLE

All game state lives on the laptop.

Tags are dumb — cheap RFID chips, UID only — never written to.

Stations are dumb — they report events and draw whatever they're told.

The bridge is dumb — just a relay between ESP-NOW and USB serial.

Reset the whole game? One function call.



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Every Station, The Same Build

ESP32

WiFi + ESP-NOW radio built in

MFRC522

Reads a tag's UID over SPI — never writes

WS2812B ×5

One wire, addressable status LEDs

+ one input

A limit switch, a joystick, or nothing

≤ 8

stations sharing one
bridge

5

LEDs per station strip

32 B

max bytes in one
ESP-NOW packet



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BUDGET

What It Costs

Component	Unit cost	Qty	Notes
ESP32 dev board	~\$6	9	8 stations + 1 bridge
MFRC522 RFID reader	~\$2	9	One per station + bridge
WS2812B strip (5px)	~\$1	8	Status LEDs per station
Limit switch / joystick	~\$1.50	6	Only input-driven stations
RFID tags (13.56 MHz)	~\$0.15	30	Ingredients + plates
Wiring, perfboard, case	~\$4	9	Rough per-station estimate

~\$130

estimated total build cost for 8 stations + bridge — *placeholder figure, swap in your real receipts*



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Six Jobs, One Shared Skeleton

Cutting Board

INPUT Limit-switch presses

RESULT Raw → Chopped

Sink

NEW

INPUT Joystick scrubbed, any direction

RESULT Configurable wash step

Frying Pan

INPUT Joystick: circle or zigzag

RESULT Raw → Cooked

Plate

INPUT None — the reader IS the plate

RESULT Collects whatever's placed on it

Pot

INPUT None — timed by the server

RESULT Raw → Cooked → Burnt

Delivery

INPUT Touch a plate's tag here

RESULT Matches an order, scores points



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Talking Over ESP-NOW

STATION → BRIDGE

- Hello, Heartbeat
- TagPlaced, TagRemoved
- TaskProgress, TaskDone

BRIDGE → STATION

- Welcome, Accept
- Reject
- SetDisplay

5× retries

200ms apart

Acked and de-duplicated by (MAC, sequence) — a lost
ack never runs a handler twice.

TX AABBCCDDEEFF 6 R 02

RX AABBCCDDEEFF 2 04DEADBEEF000000000000

ASCII lines, hex payloads — debuggable straight from a serial monitor.



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The Server Owns Every Bit Of State

Pure Engine

A single class runs on events, a tick, and an injectable clock — no serial or network inside, so tests drive it with fake time.

One Behaviour Per Kind

A small class per station kind decides the rules; the numbers live in a level file, not in code.

Calibration Wizard

Touch any tag to the bridge to name it, touch it to every station, then touch each ingredient once. Saved to disk, reloadable.

Live Web UI

Pushes the full game state over a websocket, ten times a second, to a plain-JavaScript scoreboard — no build step.



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Built To Survive Contact With Real Hardware

One Source Of Truth — a test parses the C++ header and fails the build the moment Python's protocol drifts from the firmware.

Zero-Hardware Dev — a simulated bridge speaks the real line protocol — the whole game runs with zero boards plugged in.

Tested End To End — Python tests plus native C++ unit tests for each station's pure logic — no ESP32 required.

102

Python tests,
sub-second

21

native C++ tests, three
firmware projects

0

boards needed to run
any of them



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WHAT'S NEXT

What's Cooking Next

- 01 **Tune on real hardware** — RC522 removal timing, joystick dead zone, cook times
- 02 **Give the sink a real recipe** — it's wired end to end, but no ingredient uses it yet
- 03 **Show plate contents on their LEDs** — one more display mode, mirrored on both sides
- 04 **Read more than one tag at once** — several foods on a single plate, via RC522 anticollision
- 05 **Watch bridge health** — catch a hung-but-connected bridge, not just a disconnected one



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CREDITS

The Team



[Name]

Firmware & Stations



[Name]

Game Server &
Protocol



[Name]

Hardware & Wiring



[Name]

UI & Product

[Add names, roles, and GitHub / contact links before presenting.]



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THANK YOU

Come Break It.

Jam the joystick, yank a tag mid-chop, unplug the bridge mid-round — the server is built to shrug it off and pick up right where you left it.

undercooked-irl

[Names / contact]



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